



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 1 2024/2025

LAB 2

SECR 1213 – NETWORK COMMUNICATIONS

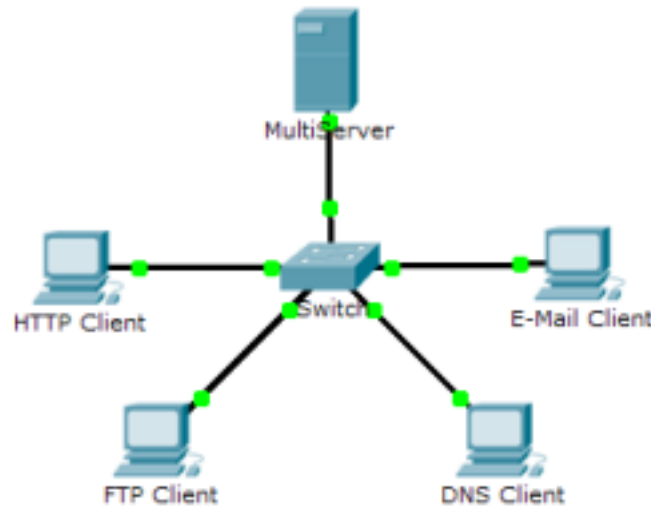
SECTION 12

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Packet Tracer Simulation - TCP and UDP Communications

Topology



Objectives

Part 1: Generate Network Traffic in Simulation Mode

Part 2: Examine the Functionality of the TCP and UDP Protocols

Background

This simulation activity is intended to provide a foundation for understanding the TCP and UDP in detail. Simulation mode provides the ability to view the functionality of the different protocols.

As data moves through the network, it is broken down into smaller pieces and identified in some fashion so that the pieces can be put back together. Each of these pieces is assigned a specific name (protocol data unit [PDU]) and associated with a specific layer. Packet Tracer Simulation mode enables the user to view each of the protocols and the associated PDU. The steps outlined below lead the user through the process of requesting services using various applications available on a client PC.

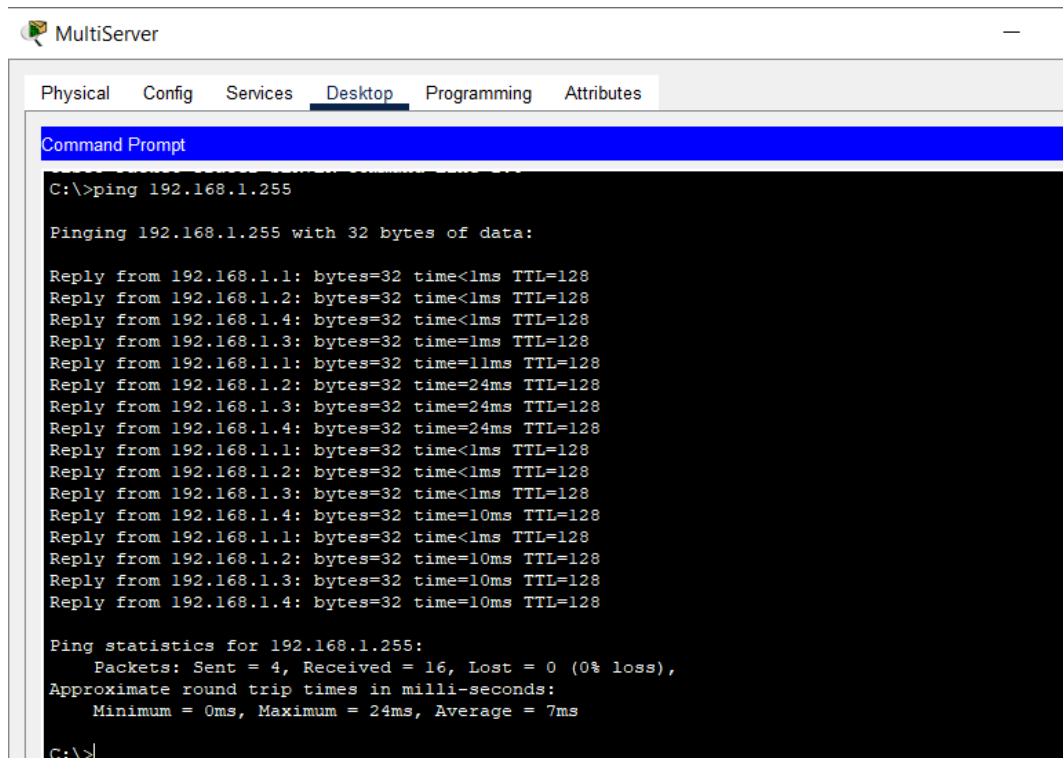
This activity provides an opportunity to explore the functionality of the TCP and UDP protocols, multiplexing and the function of port numbers in determining which local application requested the data or is sending the data.

Part 1: Generate Network Traffic in Simulation Mode

Step 1: Generate traffic to populate Address Resolution Protocol (ARP)

tables. Perform the following tasks to reduce the amount of network traffic viewed in the simulation.

- Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ping 192.168.1.255** command. This will take a few seconds as every device on the network responds to **MultiServer**.
- Close the **MultiServer** window.



The screenshot shows the MultiServer application window with the 'Desktop' tab selected. The 'Command Prompt' window is open, displaying the results of a ping command to 192.168.1.255. The output shows 4 packets sent, 16 received, and 0 lost, with round trip times ranging from 0ms to 24ms.

```

C:\>ping 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

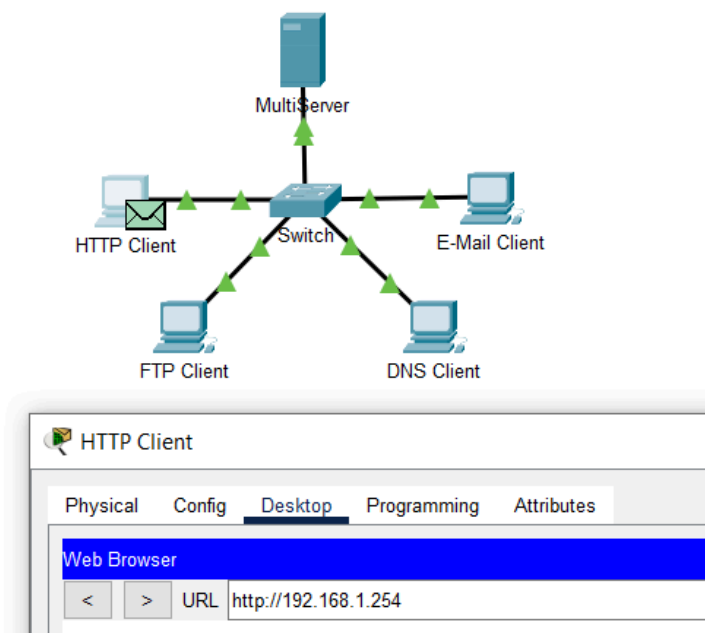
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time=24ms TTL=128
Reply from 192.168.1.3: bytes=32 time=24ms TTL=128
Reply from 192.168.1.4: bytes=32 time=24ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=10ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=10ms TTL=128
Reply from 192.168.1.3: bytes=32 time=10ms TTL=128
Reply from 192.168.1.4: bytes=32 time=10ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 7ms

C:\>
  
```

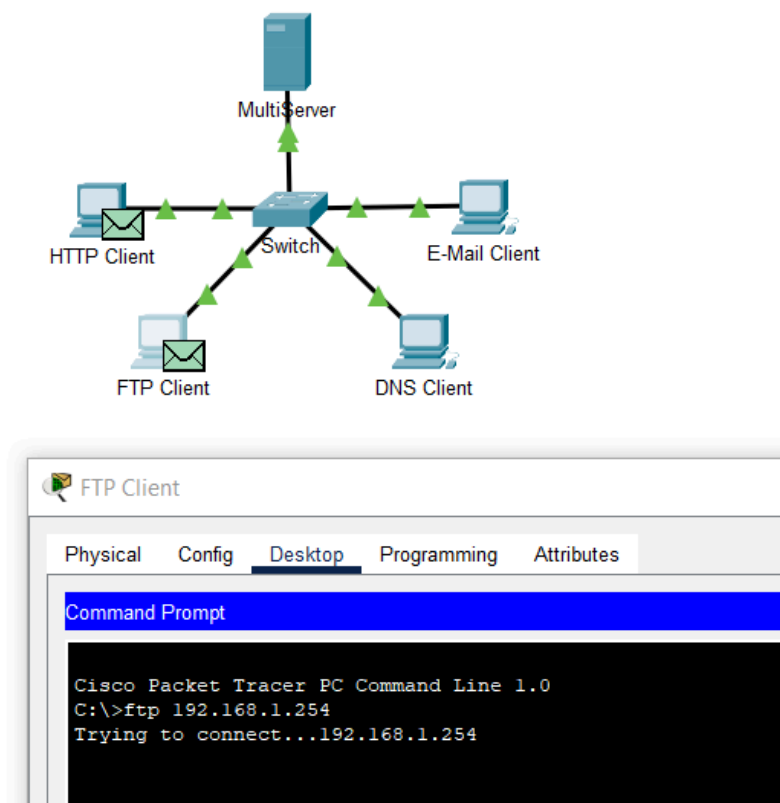
Step 2: Generate web (HTTP) traffic.

- Switch to Simulation mode.
- Click **HTTP Client** and click the **Desktop** tab > **Web Browser**.
- In the URL field, enter **192.168.1.254** and click **Go**. Envelopes (PDUs) will appear in the simulation window.
- Minimize, but do not close, the **HTTP Client** configuration window.



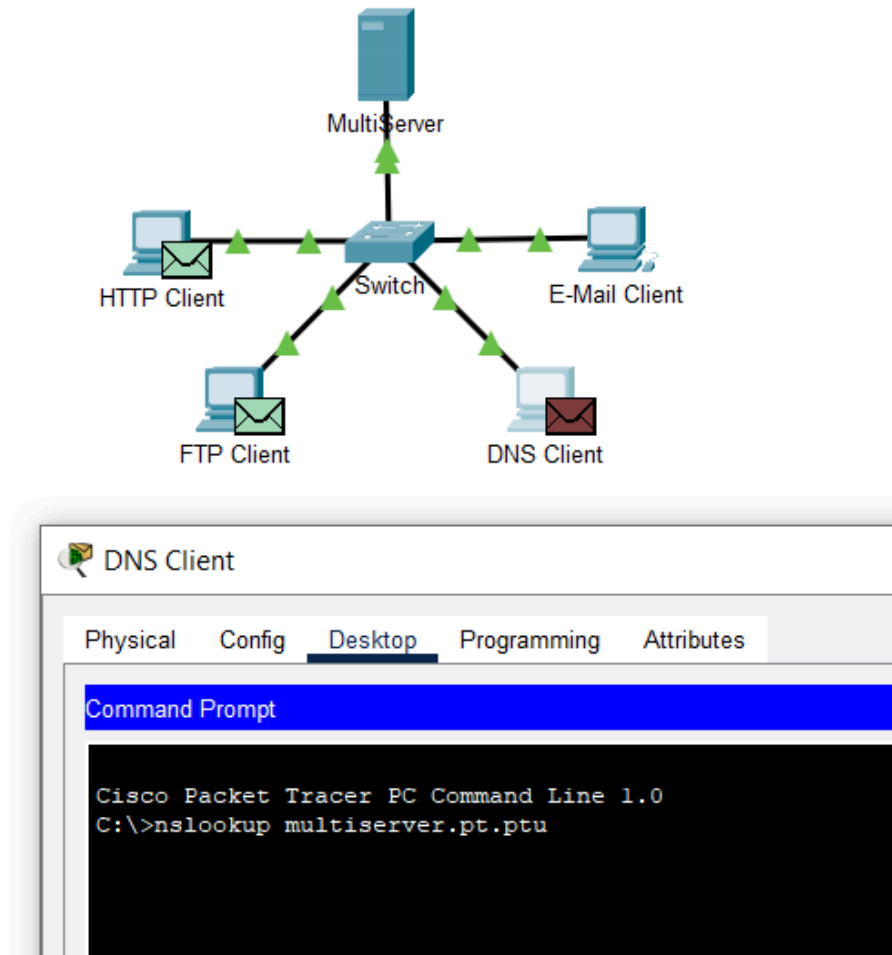
Step 3: Generate FTP traffic.

- Click **FTP Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **ftp 192.168.1.254** command. PDUs will appear in the simulation window.
- Minimize, but do not close, the **FTP Client** configuration window.



Step 4: Generate DNS traffic.

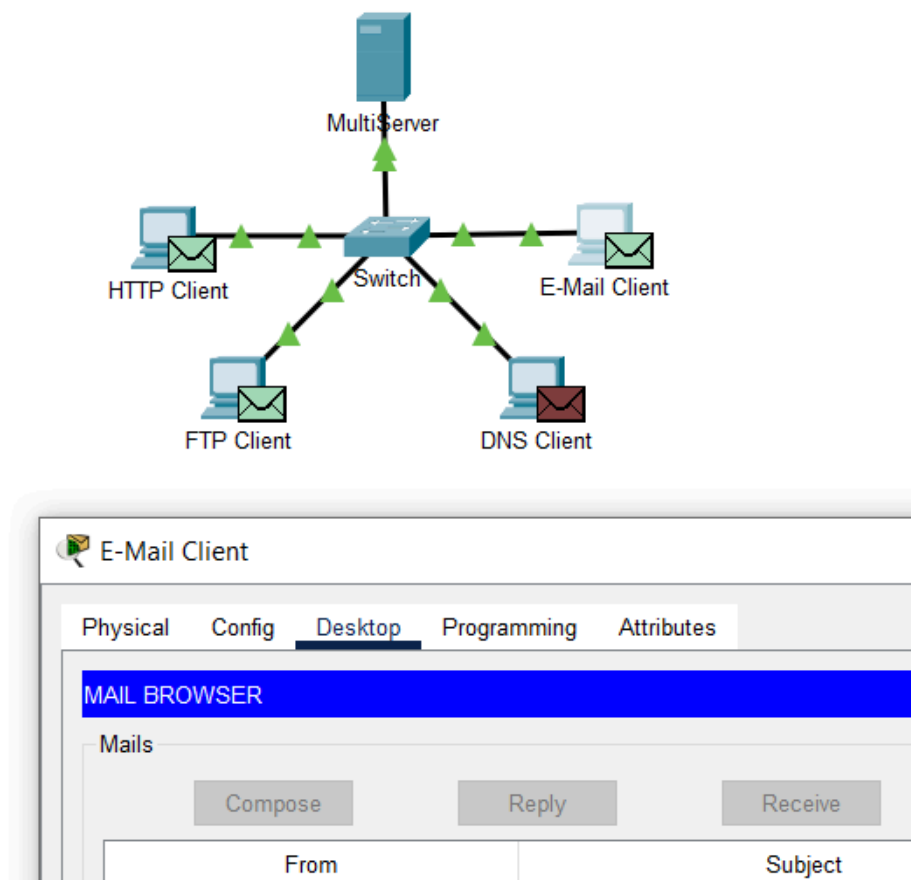
- Click **DNS Client** and click the **Desktop** tab > **Command Prompt**.
- Enter the **nslookup multiserver.pt.ptu** command. A PDU will appear in the simulation window.
- Minimize, but do not close, the **DNS Client** configuration window.



Step 5: Generate Email traffic.

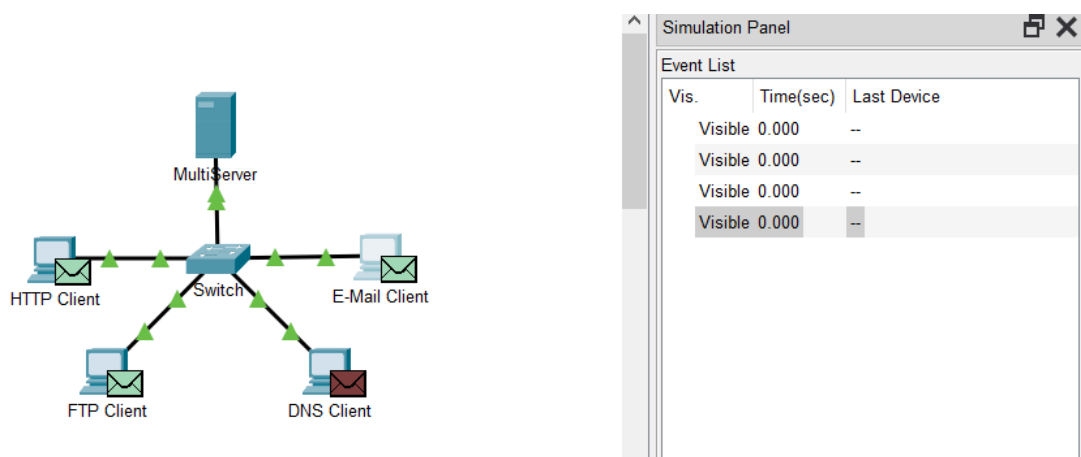
- Click **E-Mail Client** and click the **Desktop** tab > **E Mail** tool.
- Click **Compose** and enter the following information:
 - 1) **To:** user@multiserver.pt.ptu
 - 2) **Subject:** Personalize the subject line
 - 3) **E-Mail Body:** Personalize the Email
- Click **Send**.
- Minimize, but do not close, the **E-Mail Client** configuration window.

Packet Tracer Simulation - TCP and UDP Communications



Step 6: Verify that the traffic is generated and ready for simulation.

Every client computer should have PDUs listed in the Simulation Panel.

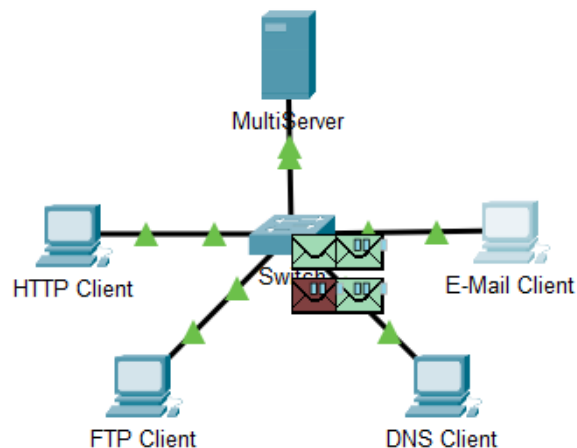


Part 2: Examine Functionality of the TCP and UDP Protocols

Step 1: Examine multiplexing as all of the traffic crosses the network.

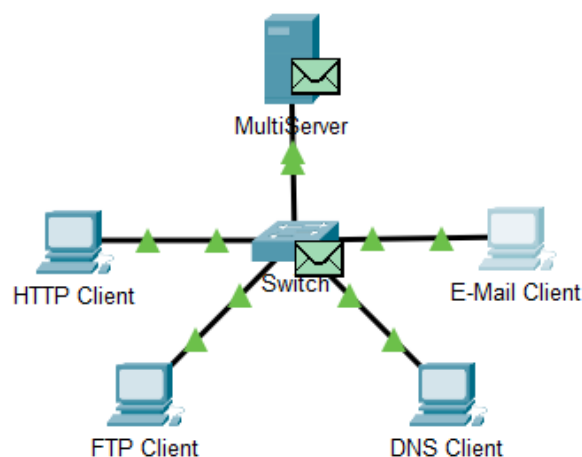
You will now use the **Capture/Forward** button and the **Back** button in the Simulation Panel.

- a. Click **Capture/Forward** once. All of the PDUs are transferred to the switch.



- b. Click **Capture/Forward** again. Some of the PDUs disappear. What do you think happened to them?

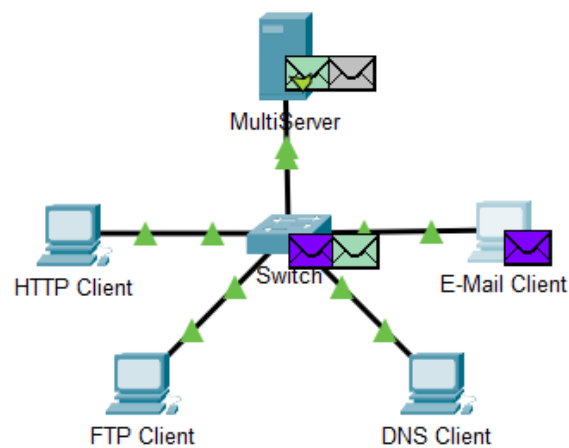
Ans: The disappearing PDUs have been stored in the switch.



- c. Click **Capture/Forward** six times. All clients should have received a reply. Note that only one PDU can cross a wire in each direction at any given time. What is this called?

Ans: Multiplexing

Packet Tracer Simulation - TCP and UDP Communications



- d. A variety of PDUs appears in the event list in the upper right pane of the simulation window. Why are there so many different colors?

Ans: The different colors represented as different protocols.

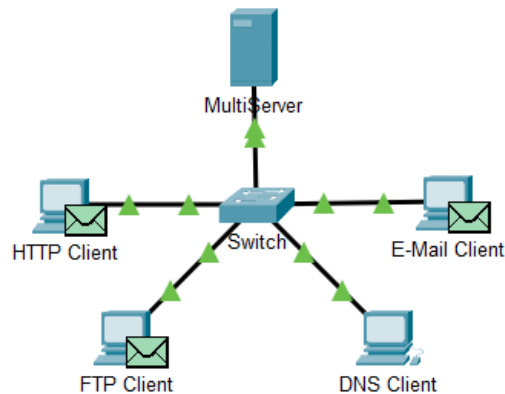
Simulation Panel		
Event List		
Vis.	Time(sec)	Last Device
	0.006	FTP Client
	0.006	Switch
	0.007	Switch
	0.007	Switch
	0.007	--
	0.007	--
Visible	0.008	Switch
Visible	0.008	--
Visible	0.008	MultiServer
Visible	0.008	E-Mail Client
Visible	0.008	--
	0.009	E-Mail Client
	0.009	MultiServer
	0.009	Switch
	0.009	Switch
	0.009	--

- e. Click **Back** eight times. This should reset the simulation.

Note: Do not click **Reset Simulation** any time during this activity; if you do, you will need to repeat the steps in Part 1.

Step 2: Examine HTTP traffic as the clients communicate with the server.

- a. Filter the traffic that is currently displayed to display only **HTTP** and **TCP** PDUs filter the traffic that is currently displayed:
 - 1) Click **Edit Filters** and toggle the **Show All/None** checkbox.
 - 2) Select **HTTP** and **TCP**. Click anywhere outside of the Edit Filters box to hide it. The Visible Events should now display only **HTTP** and **TCP** PDUs.



- b. Click **Capture/Forward**. Hold your mouse above each PDU until you find one that originates from **HTTP Client**. Click the PDU envelope to open it.
- c. Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled? Are these communications considered to be reliable?

Ans: The section labeled is TCP. Yes, these communications are considered to be reliable.

PDU Information at Device: Switch

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

DST IP:192.168.1.254			
DATA (VARIABLE LENGTH)			
TCP			
SOURCE PORT:1025		DESTINATION PORT:80	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET:0x0	RESERVED:0	FLAGS:0b00000010	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

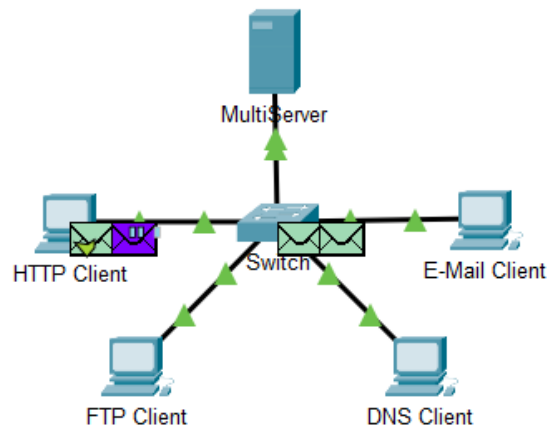
Packet Tracer Simulation - TCP and UDP Communications

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 80, 0, 0. SYN

- SRC PORT: 1025
- DEST PORT: 80
- SEQUENCE NUM: 0
- ACK NUM: 0

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **HTTP Client** with a checkmark.



- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 80, 1025, 0, 1. SYN+ACK.

- SRC PORT: 80
- DEST PORT: 1025
- SEQUENCE NUM: 0
- ACK NUM: 1

Compared to before, the source and destination ports are reversed, and the acknowledgement number is 1. The SYN also has changed to SYN+ACK.

Packet Tracer Simulation - TCP and UDP Communications

PDU Information at Device: HTTP Client

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

SRC IP:192.168.1.254			
DST IP:192.168.1.1			
DATA (VARIABLE LENGTH)			

TCP

0		4		8		16		24		Bits	
SOURCE PORT:80						DESTINATION PORT:1025					
SEQUENCE NUMBER:0											
ACKNOWLEDGEMENT NUMBER:1											
OFFSET:0x0		RESERVED:0		FLAGS:0b00010010				WINDOW:16384			
CHECKSUM:0x0000						URGENT POINTER:0x0000					
OPTION											
DATA (VARIABLE LENGTH)										PADDING: 0	

- g. There is a second **PDU** of a different color, which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.

PDU Information at Device: HTTP Client

OSI Model Outbound PDU Details

PDU Formats

DATA (VARIABLE LENGTH)			
------------------------	--	--	--

TCP

0		4		8		16		24		Bits	
SOURCE PORT:1025						DESTINATION PORT:80					
SEQUENCE NUMBER:1											
ACKNOWLEDGEMENT NUMBER:1											
OFFSET:0x0		RESERVED:0		FLAGS:0b00011000				WINDOW:65535			
CHECKSUM:0x0000						URGENT POINTER:0x0000					
OPTION											
DATA (VARIABLE LENGTH)										PADDING: 0	

HTTP REQUEST

0				4				8				16				Bytes			
HTTP Data:Accept-Language: en-us Accept: */*																			

Packet Tracer Simulation - TCP and UDP Communications

- h. What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Ans: 1025, 80, 1, 1. PSH+ACK.

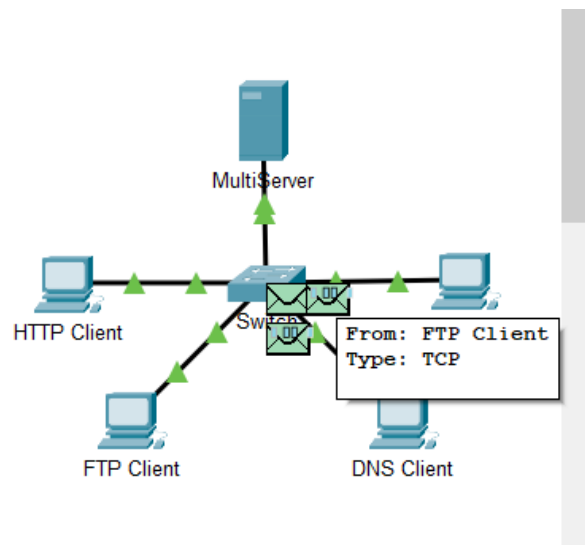
- SRC PORT: 1025
- DEST PORT: 80
- SEQUENCE NUM: 1
- ACK NUM: 1

Then, the source and destination ports are reversed. Both sequence and acknowledgement numbers are 1.

- i. Click **Back** until the simulation is reset.

Step 3: Examine FTP traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **FTP Client**. Click that PDU envelope to open it.



- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled? Are these communications considered to be reliable?

Ans: TCP. Yes these communications are considered to be reliable.

PDU Information at Device: Switch

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

DST IP:192.168.1.254			
DATA (VARIABLE LENGTH)			
TCP			
0 4 8 16 24 Bits			
SOURCE PORT:1025		DESTINATION PORT:21	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:0			
OFFSET:0x0	RESERVED:0	FLAGS:0b00000010	WINDOW:65535
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

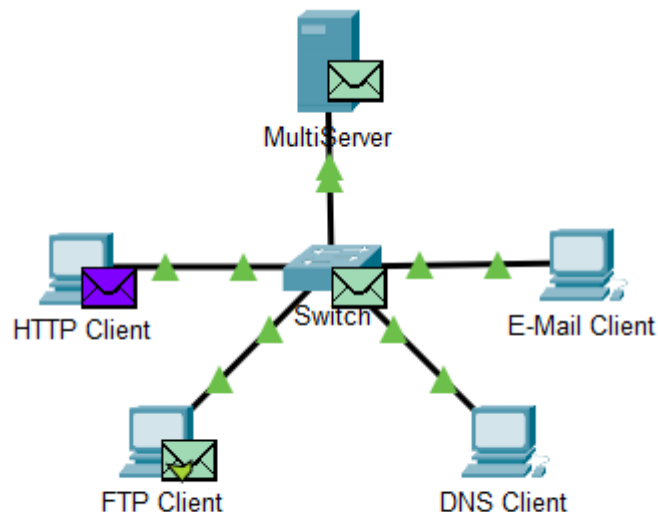
Packet Tracer Simulation - TCP and UDP Communications

- d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 21, 0, 0. SYN

- SRC PORT: 1025
- DEST PORT: 21
- SEQUENCE NUM: 0
- ACK NUM: 0

- e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.



- f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 21, 1025, 0, 1. SYN+ACK.

- SRC PORT: 21
- DEST PORT: 1025
- SEQUENCE NUM: 0
- ACK NUM: 1

The source and destination ports are reversed. The acknowledgement number is 1.

PDU Information at Device: FTP Client

OSI Model [Inbound PDU Details](#) Outbound PDU Details

PDU Formats

SRC IP:192.168.1.254			
DST IP:192.168.1.2			
DATA (VARIABLE LENGTH)			
TCP			
SOURCE PORT:21		DESTINATION PORT:1025	
SEQUENCE NUMBER:0			
ACKNOWLEDGEMENT NUMBER:1			
OFFSET:0x0	RESERVED:0	FLAGS:0b00010010	WINDOW:16384
CHECKSUM:0x0000		URGENT POINTER:0x0000	
OPTION			
DATA (VARIABLE LENGTH)			PADDING: 0

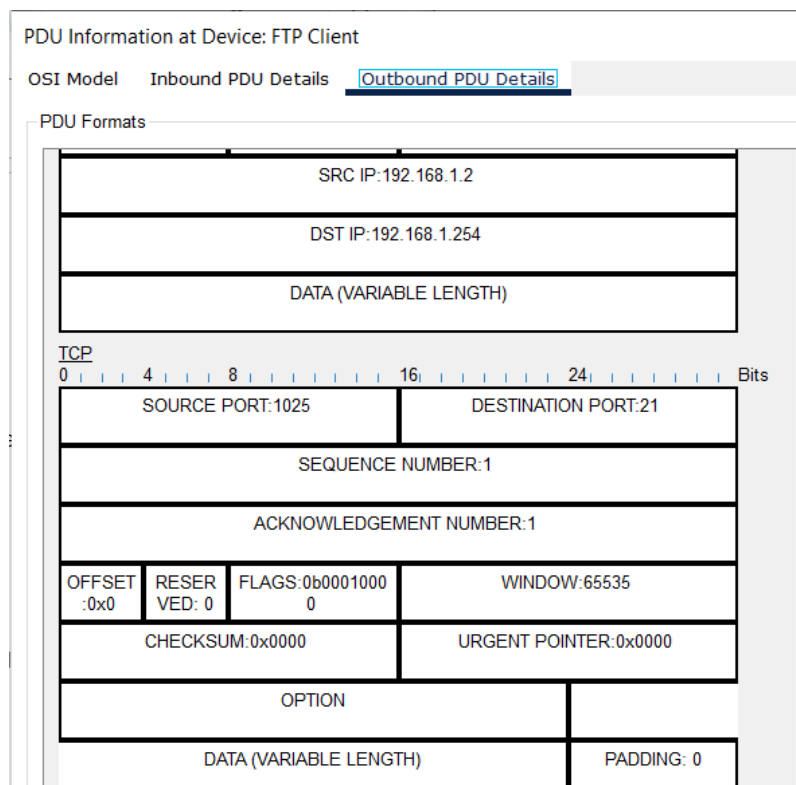
Packet Tracer Simulation - TCP and UDP Communications

- g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

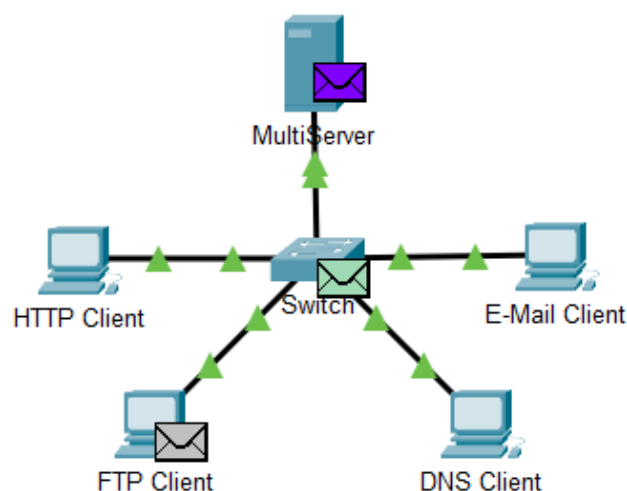
Ans: 1025, 21, 1, 1. ACK

- SRC PORT: 1025
- DEST PORT: 21
- SEQUENCE NUM: 1
- ACK NUM: 1

The source and destination ports are reversed. Both sequence and acknowledgement numbers are 1.



- h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.



Packet Tracer Simulation - TCP and UDP Communications

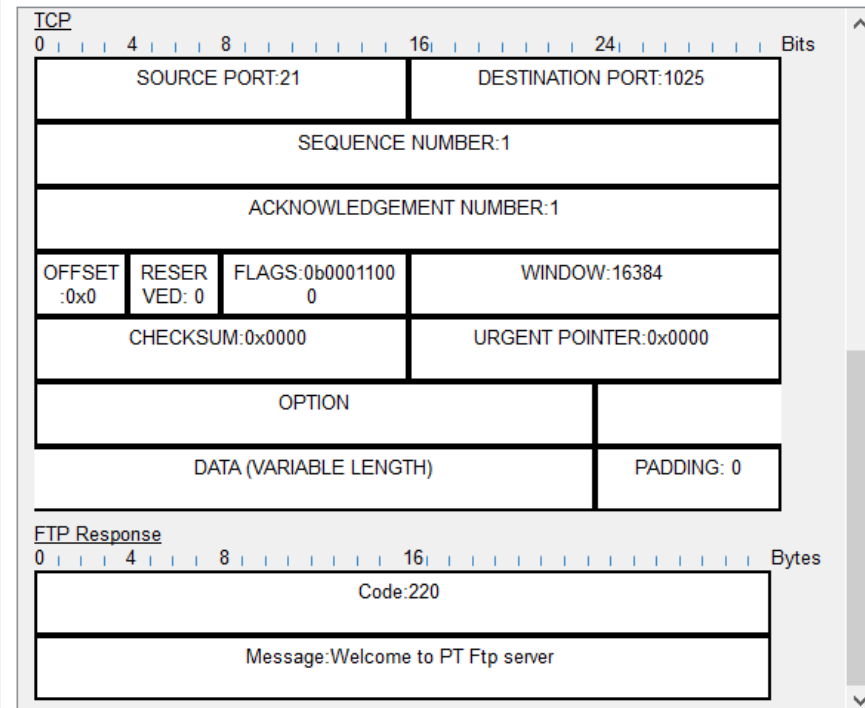
- i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section. What is the message from the server?

Ans: The message might say "Welcome to PT Ftp server".

PDU Information at Device: FTP Client

OSI Model Inbound PDU Details

PDU Formats

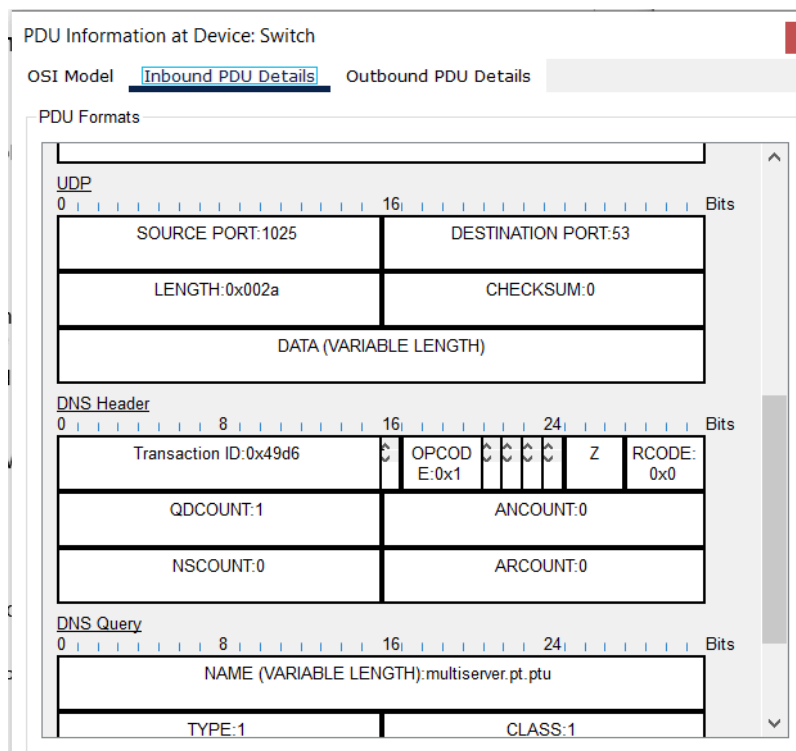


- j. Click **Back** until the simulation is reset.

Step 4: Examine DNS traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.
- Click the PDU envelope to open it.
- Click the **Inbound PDU Details** tab and scroll down to the last section. What is the section labeled? Are these communications considered to be reliable?

Ans: UDP. These communications are not considered to be reliable.



- Record the **SRC PORT** and **DEST PORT** values. Why is there no sequence and acknowledgement number?

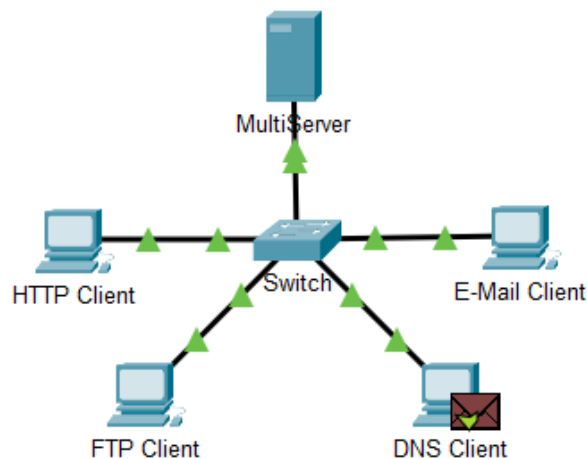
Ans:

Source Port: 1025

Destination Port: 53.

Because UDP does not need to establish a reliable connection.

- Close the **PDU** and click **Capture/Forward** until a PDU returns to the **DNS Client** with a checkmark.



f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans:

Source Port: 53

Destination Port: 1025

The source and destination ports are reversed.

PDU Information at Device: DNS Client

OSI Model [Inbound PDU Details](#)

PDU Formats

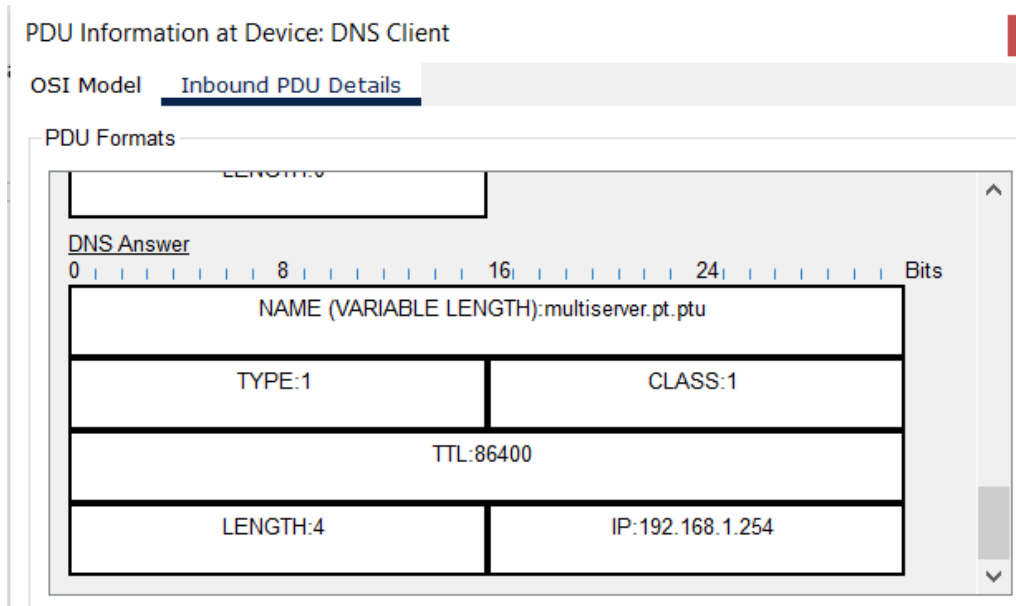
UDP																															
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 Bits																															
SOURCE PORT:53																DESTINATION PORT:1025															
LENGTH:0x004a																CHECKSUM:0															
DATA (VARIABLE LENGTH)																															

DNS Header																															
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 Bits																															
Transaction ID:0x49d6																OPCODE E:0x1		Z		RCODE: 0x0											
QDCOUNT:1																ANCOUNT:1															
NSCOUNT:0																ARCOUNT:0															

DNS Query																															
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 Bits																															
NAME (VARIABLE LENGTH):multiserver.pt.ptu																															
TYPE:1																CLASS:1															

g. What is the last section of the **PDU** called?

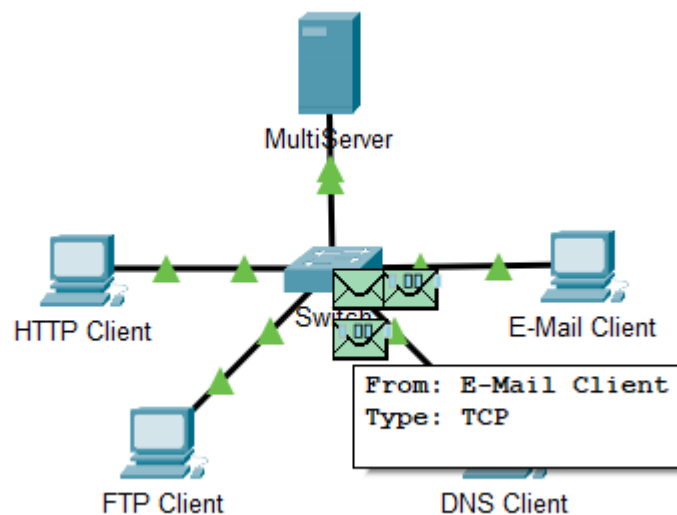
Ans: DNS ANSWER.



h. Click **Back** until the simulation is reset.

Step 5: Examine email traffic as the clients communicate with the server.

- In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**.
- Click **Capture/Forward**. Hold your cursor above each PDU until you find one that originates from **E-mail Client**. Click that PDU envelope to open it.



- Click the **Inbound PDU Details** tab and scroll down to the last section. What transport layer protocol does email traffic use? Are these communications considered to be reliable?

Ans: TCP. Yes, these communications are considered to be reliable.

PDU Information at Device: Switch

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

TCP					
0	4	8	16	24	Bits
SOURCE PORT:1025		DESTINATION PORT:25			
SEQUENCE NUMBER:0					
ACKNOWLEDGEMENT NUMBER:0					
OFFSET:0x0	RESERVED: 0	FLAGS:0b00000010	WINDOW:65535		
CHECKSUM:0x0000			URGENT POINTER:0x0000		
OPTION					
DATA (VARIABLE LENGTH)					PADDING: 0

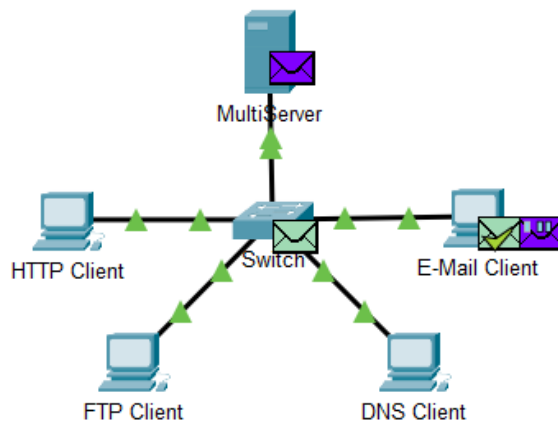
Packet Tracer Simulation - TCP and UDP Communications

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is written in the field to the left of the **WINDOW** field?

Ans: 1025, 25, 0, 0. SYN.

- SRC PORT: 1025
- DEST PORT: 25
- SEQUENCE NUM: 0
- ACK NUM: 0

e. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.



f. Click the PDU envelope and select **Inbound PDU Details**. How are the port and sequence numbers different than before?

Ans: 25, 1025, 0, 1. SYN+ACK.

- SRC PORT: 25
- DEST PORT: 1025
- SEQUENCE NUM: 0
- ACK NUM: 1

The source and destination ports are reversed. The acknowledgement number is 1.

PDU Information at Device: E-Mail Client

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

TCP															
0	4	8	16	24	Bits										
SOURCE PORT:25								DESTINATION PORT:1025							
SEQUENCE NUMBER:0															
ACKNOWLEDGEMENT NUMBER:1															
OFFSET :0x0				RESER VED: 0		FLAGS:0b0001001 0				WINDOW:16384					
CHECKSUM:0x0000								URGENT POINTER:0x0000							
OPTION															
DATA (VARIABLE LENGTH)												PADDING: 0			

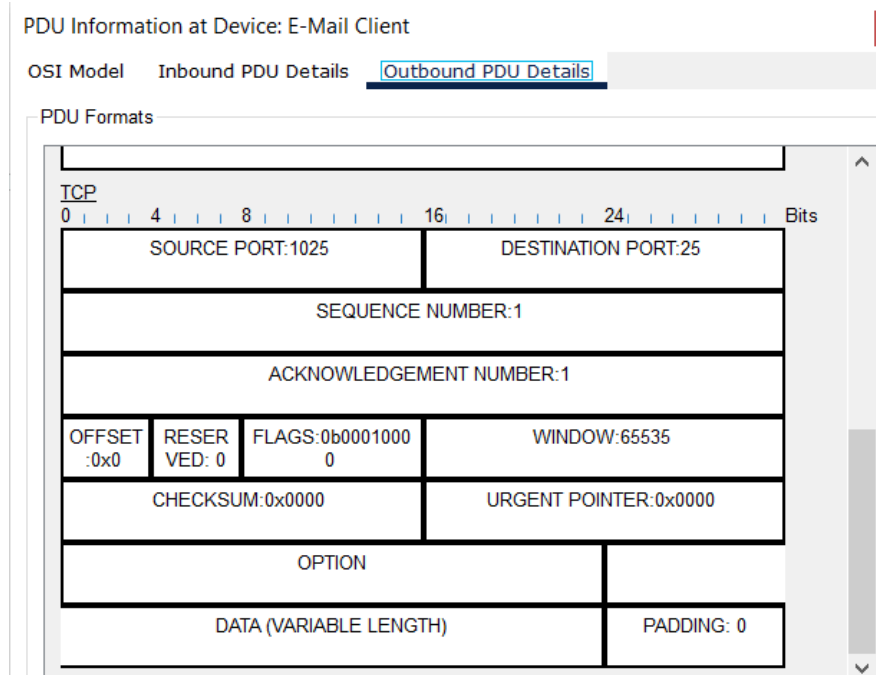
Packet Tracer Simulation - TCP and UDP Communications

g. Click the **Outbound PDU Details** tab. How are the port and sequence numbers different from the previous two results?

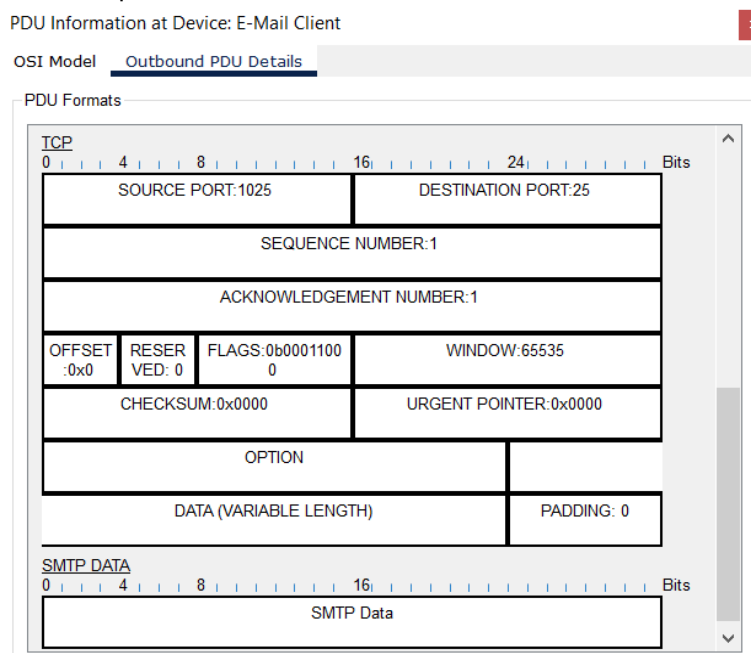
Ans:25, 1025, 0, 1. SYN+ACK.

- SRC PORT: 1025
- DEST PORT: 25
- SEQUENCE NUM: 1
- ACK NUM: 1

The source and destination ports are reversed. The sequence and acknowledgement number is 1.



h. There is a second **PDU** of a different color that **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.



i. How are the port and sequence numbers different from the previous two **PDU**s?

Ans: 1025, 25, 1, 1. PSH+ACK.

- SRC PORT: 1025
- DEST PORT: 25
- SEQUENCE NUM: 1
- ACK NUM: 1

The source and destination ports are reversed. Both sequence and acknowledgement numbers are 1.

j. What email protocol is associated with TCP port 25? What protocol is associated with TCP port 110?

Ans: SMTP and POP3 protocol.

k. Click **Back** until the simulation is reset.

Step 6: Examine the use of port numbers from the server.

a. To see TCP active sessions, perform the following steps in quick succession:

- 1) Switch back to **Realtime** mode.
- 2) Click **MultiServer** and click the **Desktop** tab > **Command Prompt**.

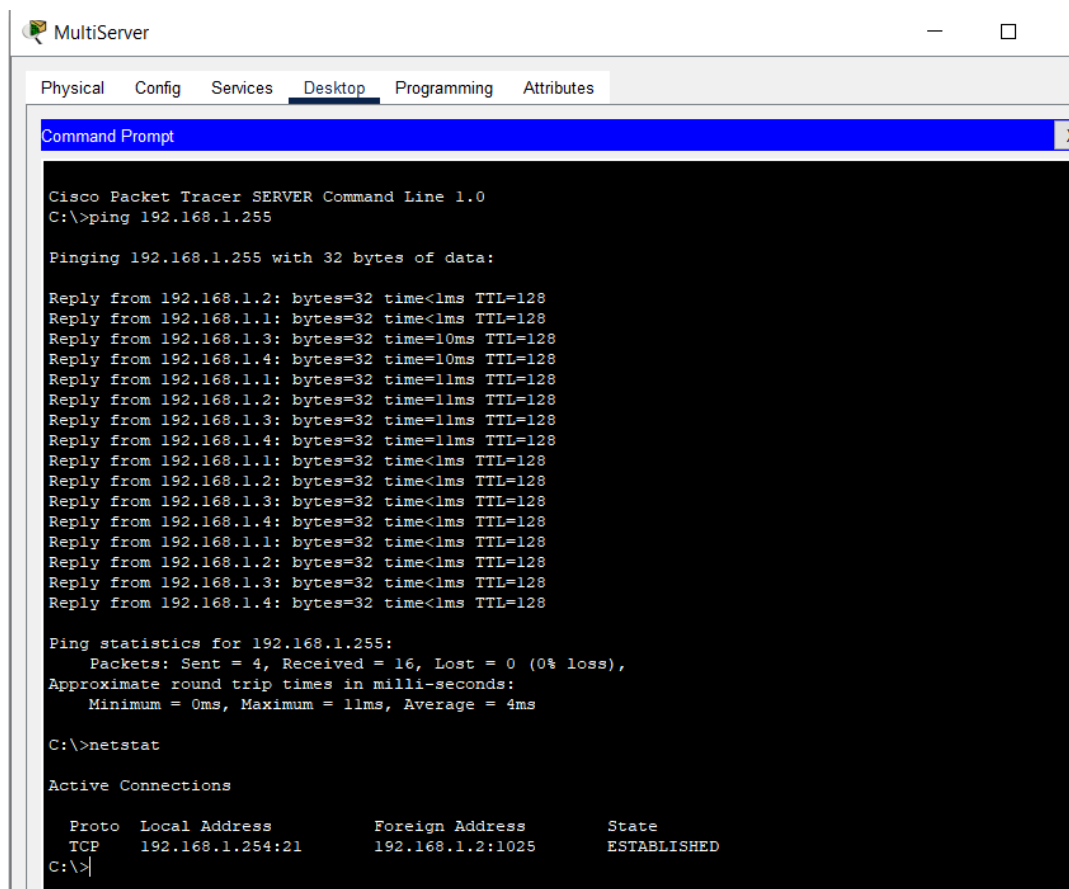
b. Enter the **netstat** command. What protocols are listed in the left column?

What port numbers are being used by the server?

Ans:

Port numbers: 21

Protocols: TCP



```

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.1.255

Pinging 192.168.1.255 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=10ms TTL=128
Reply from 192.168.1.4: bytes=32 time=10ms TTL=128
Reply from 192.168.1.1: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.3: bytes=32 time=11ms TTL=128
Reply from 192.168.1.4: bytes=32 time=11ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.255:
    Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>netstat

Active Connections

   Proto Local Address          Foreign Address         State
   TCP    192.168.1.254:21       192.168.1.2:1025       ESTABLISHED
C:\>
  
```

c. What states are the sessions in?

Ans: ESTABLISHED.

d. Repeat the **netstat** command several times until you see only one session still ESTABLISHED.
For which service is this connection still open?

Ans: FTP.

Packet Tracer Simulation - TCP and UDP Communications

```
Ping statistics for 192.168.1.255:
  Packets: Sent = 4, Received = 16, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 11ms, Average = 4ms

C:\>netstat

Active Connections

   Proto Local Address           Foreign Address         State
   TCP    192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>netstat

Active Connections

   Proto Local Address           Foreign Address         State
   TCP    192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>netstat

Active Connections

   Proto Local Address           Foreign Address         State
   TCP    192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
C:\>netstat

Active Connections

   Proto Local Address           Foreign Address         State
   TCP    192.168.1.254:21        192.168.1.2:1025       ESTABLISHED
```

Why doesn't this session close like the other three? (Hint: Check the minimized clients)

Ans: The server is waiting for a password from the client.

Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 2: Examine Functionality of the TCP and UDP Protocols	Step 1	15	
	Step 2	15	
	Step 3	15	
	Step 4	15	
	Step 5	15	
	Step 6	25	
Total Score		100	